Muon Neutrino Charged Current Single Pion Cross Section on Argon using Run 1 NuMI Data

Internal Note — Version 0.3 (DRAFT)

DRAFT STATUS — READ FIRST

This note mirrors the structure of the BNB CC1π internal note (`BNBInternalNoteCC1pi_V0.91.pdf`, J. P. Detje, 4 Feb 2025) for the NuMI Run 1 FHC measurement.

Sections 1–5 describe the analysis ***as it is actually configured in the code today*** *and every number in them was read from the source or configs. Section 6 (fake-data tests) now carries* ***real results*** *from the re-run chain: the GENIE closure χ^2 for all five observables and the four-generator comparison on the corrected flux. Section 7 (real-data results) is* ***still empty*** *— the real beam-on sample has not been restored and the chain has not been run on data (see §2.2 and §7).*

The single most important change since v0.1 is the ***flux-normalisation correction*** *(§2.4): the NuMI flux constant was ~3.48× too large, which had made every extracted cross section ~3.5× too small. Every number in §6 uses the corrected flux. Nothing in §7 should be quoted as a data result.*

Changes in v0.3: a full extraction-bias investigation (§6.7) — settled as ***no framework unfolding bug*** *(the ~1.3× was the fake-data POT offset), confirmed by a numuMC self-closure and an independent cross-pipeline validation; a detailed selection cut table (§3.4); response matrices and efficiencies for all five observables (§3.6–3.7); the systematic covariance-matrix breakdown (§4.7); and the muon BDT is now active in the selection (§8.4).*

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1 Overview

This note describes the extraction of the $\nu\mu$ charged-current single charged pion cross section on argon using MicroBooNE Run 1 FHC data from the NuMI beam. It is the NuMI counterpart of the BNB analysis of the same final state, and is intended to be read alongside that note, which contains fuller descriptions of the selection variables and the boosted decision trees.

The measurement is performed with the common `xsec_analyzer` framework: event selection via the `CC1mu1piXp` selection class, universe construction via `univmake`, and unfolding via the Wiener-SVD implementation in `CrossSectionExtractor`. Section 2 describes the software and samples, Section 3 the signal definition and selection, Section 4 the uncertainty treatment, Section 5 the cross-section extraction and unfolding, Section 6 the fake-data tests, and Section 7 the results.

The principal differences from the BNB analysis are:

	BNB analysis	**This analysis**
Beam	BNB	NuMI FHC
Runs	1–5	1 only
Flux systematic	`weight_flux_all`	`weight_ppfx_all` (PPFX)
Beamline geometry systematic	n/a	NuMI-specific — **not currently included**, see §4.6
Proton multiplicity	—	Xp (any number, no threshold)
Unfolding	Wiener-SVD	Wiener-SVD (D'Agostini available as a cross-check)

2 Software and Samples

2.1 Framework

The analysis runs entirely within `xsec_analyzer`. The chain is:

1. `ProcessNTuples` — applies the `CC1mu1piXp` selection to raw PeLEE ntuples and writes `stv_tree` output (`run_process_ntuples.sh`).
2. `univmake` — builds systematic universe histograms from a bin config (`run_universe_maker.sh`).
3. `Unfolder` / `UnfolderNuMI` — extracts and unfolds the cross section (`run_unfolder.sh`).

2.2 Samples

Run 1 FHC, as configured in `configs/file_properties_numi.txt`:

Sample type	**Purpose**
`onBNB`	beam-on data (see warning below)
`extBNB`	beam-off (EXT), 3 821 593 triggers
`numuMC`	NuMI overlay CV
`dirtMC`	dirt overlay
`detVarCV` + 8 variations	detector systematics

> ***The `onBNB` slot currently holds GENIE fake data.*** *It is the detector-variation central-value sample (a GENIE G18_10a production), symlinked as `xsec-ana-genie_detvarCV_fakedata_run1_fhc.root` and entered with its native POT (7.631×10^{20}) and beam-on-equivalent triggers (18 153 256). This sample is statistically **independent** of the numuMC overlay that builds the response matrix, so it is a genuine same-generator closure rather than a trivial self-closure. It replaced the earlier NuWro overlay fake data (`xsec-ana-numi_nuwro_overlay_pion_ntuples_run1_fhc.root`, 6.65×10^{20} POT), which carried a real cos θ_μ* shape *difference versus any standalone NuWro (§6.4).*
>
> *The real beam-on sample (`xsec-ana-neutrinoselection_filt_run1_beamon_beamgood.root`, 3.283×10^{20} POT, 7 809 962 triggers) remains **commented out** at `configs/file_properties_numi.txt`.*

*The analysis is therefore configured as a **fake-data study**, not a real-data measurement. Section 7 cannot be filled until this is swapped back and the chain re-run on data.*

Exposure and normalisation (`file_properties_numi.txt`):

Quantity	Value
Real beam-on POT (commented out)	3.283×10^{20} (7 809 962 triggers)
GENIE detVar-CV fake data (active onBNB)	7.631×10^{20} (18 153 256 equiv. triggers)
Retired NuWro fake data	6.650×10^{20}
Beam-off (EXT) triggers	3 821 593
MC → data POT scale	run POT / summed MC POT (≈ 0.141)
Integrated $\nu\mu+\bar{\nu}\mu$ flux Φ	6.81159×10^{-10} cm^{-2}/POT (E > 60 MeV, §2.4)
$\bar{\nu}\mu$ fraction of flux	≈ 37.8 %

All MC, EXT and dirt samples are POT-scaled to the active onBNB exposure; the fake data enters at its native POT. Because the flux is a pure normalisation divisor, the corrected Φ changes only the absolute scale, not the shapes.

2.3 Detector variation samples

Eight variations are used: `detVarLYdown`, `detVarLYrayl`, `detVarRecomb2`, `detVarSCE`, `detVarWMAngleXZ`, `detVarWMAngleYZ`, `detVarWMX`, `detVarWMYZ`.

The BNB note additionally lists `detVarLYatten`; it is not present in the NuMI sample set here. Like the BNB analysis, the WireMod dE/dx variation is not used, in favour of the recombination variation.

2.4 Corrections applied since the previous iteration

The following were found and fixed during a code review of this analysis. All of them affect the numbers a run would produce, which is why no results from before them should be carried forward:

- **Flux normalisation (dominant correction).** The integrated NuMI flux constant in the cross-section conversion factor was 2.372512×10^{-9} $\nu\mu/cm^2$/POT, taken from the superseded `uboone_numi_flux_histograms.root` (which included an unphysical ~25–30 MeV artifact spike). The authoritative value from the flux-file author, using the new GEANT4 flux integrated above 60 MeV, is 4.43515×10^{-10} ($\nu\mu$) + 2.37644×10^{-10} ($\bar{\nu}\mu$) = 6.81159×10^{-10} $\nu\mu/cm^2$/POT — a factor **3.48 smaller**. Because the cross section divides by the flux, the old constant made every extracted cross section ~3.5× too *small*. The correction was confirmed four independent ways: standalone NuWro (2.7× low) and GENIE (3.2× low) run on the old flux; the flux author's numbers (3.48×); and a GENIE closure in which the framework tune scaled by 3.48× matches a standalone GENIE on the corrected flux to 3%. Fixed in `FiducialVolume.hh` (`integrated_numu_flux_in_FV`). The flux is the active-volume-averaged flux used as an approximation for the fiducial-volume flux — a documented ~few-% residual; no fiducial-volume correction is applied (the target count is done separately in the same conversion factor).

- **Muon momentum estimator.** The estimator combining range (contained) and MCS (uncontained) was computed but never written to the output tree, so the unfolding binned on MCS alone. It is now branched as `candidate_muon_mom_reco` and `ccpi_pmu_bin_config.txt` uses it.
- **Pion momentum estimator.** Reco pion momentum was a proton-hypothesis range *kinetic energy* binned against a true *momentum* on identical bin edges. It is now the muon-hypothesis range momentum.
- **Per-category systematics.** Six `MCFullCorrCategory` entries evaluated to identically zero and have been removed; see §4.6.
- **Background true bins.** `MakeConfig` had stopped emitting them, so any regenerated bin config would have had no background subtraction at all.
- **Diagnostic histograms** are now filled with the CV weight (`spline_weight_ × tuned_cv_weight_`).

3 Signal and Selection

3.1 Signal definition

An event is signal if, in truth:

- the neutrino vertex is inside the fiducial volume: $10 < x < 246$ cm, $-101 < y < 101$ cm, $10 < z < 986$ cm;
- the interaction is charged-current;
- the neutrino is $\nu\mu$ (`|mc_nu_pdg| == 14`, see §8.3);
- there is exactly one muon above threshold;
- there is exactly one charged pion;
- there are no neutral pions, no kaons and no heavier mesons;
- any number of protons is allowed, with no momentum threshold (the ”Xp” in `CC1mu1piXp`);

and the event falls in the measured phase space:

- $p\_\mu > 0.15$ GeV/c
- $p\_\pi > 0.175$ GeV/c
- $\theta\_\mu\pi < 2.6$ rad

The phase-space cuts are applied identically in `define_signal()` and `categorize_event()`, so signal and category assignment cannot disagree. The counting of ”one muon / one charged pion” uses no momentum threshold (any `|p| > 0`); the momentum thresholds enter only through the measured phase space above.

Threshold validation. The two momentum thresholds sit at the selection efficiency turn-on, measured against a **threshold-free** true signal (the CC $\nu\mu$ $1\mu1\pi\pm$ topology with the momentum/angle cuts removed):

	below threshold	above threshold	turn-on
$p_\mu > 0.15$	~0–1.5 %	~9–12 %	0.145 → 0.155 GeV/c
$p_\pi > 0.175$	~0–3 %	~12–16 %	0.165 → 0.185 GeV/c

Both thresholds are therefore **correctly placed** — lowering them would add near-zero-efficiency phase space, raising them would discard measurable signal. The odd value 0.175 is precisely the pion turn-on knee. The muon momentum (range/MCS, §2.4) is essentially unbiased near threshold.

Pion momentum bias (open item). The pion momentum is reconstructed from track range under the **muon hypothesis** (`track_range_mom_mu`, the only range-momentum branch available). It is biased low for pions and increasingly so with momentum — mean(reco − true) grows from ≈ 0 at threshold to −0.05 at 0.25 and **−0.15 GeV/c (~40 %) at 0.40** — from two effects: the μ-vs-π mass hypothesis (~−0.025) and, dominantly, **pion hadronic re-interaction** shortening the range, which effectively saturates the reconstructed momentum near 0.2–0.25 GeV/c. The p_π response is therefore *shifted* (not merely smeared) above $p_\pi \approx$ 0.3 GeV/c, which is why the p_π binning above this must be coarse (§3.8). A pion-hypothesis range correction is recommended (§8.7).

3.2 Binning

Five differential observables are measured. Bin edges as configured:

Observable	Config	Bins (true/reco)	Edges
p_μ	`ccpi_pmu_bin_config.txt`	23/22	0.15, 0.20, 0.30 … 2.00, 2.30, 2.60, 3.00 GeV/c
p_π	`ccpi_ppi_bin_config.txt`	6/5	0.175, 0.20, 0.30, 0.40, 0.50, 1.00 GeV/c
$\cos\theta_\mu$	`ccpi_costhmu_bin_config.txt`	13/12	−1.0, −0.5, 0.0, 0.2, 0.4, 0.55, 0.65, 0.75, 0.82, 0.88, 0.93, 0.97, 1.0
$\cos\theta_\pi$	`ccpi_costhpi_bin_config.txt`	13/12	−1.0, −0.7, −0.4, −0.2, 0.0, 0.2, 0.35, 0.5, 0.65, 0.78, 0.88, 0.95, 1.0
$\theta_{\mu\pi}$	`ccpi_thmupi_bin_config.txt`	10/9	0.0 … 2.513, 2.600 rad

Each true-bin count is one greater than the reco count: the extra true bin is the background bin, defined by inverting the signal definition (`!CC1mu1piXp_MC_Signal`). All configs are single-block with no sideband reco bins.

3.3 Particle identification

Three BDTs are used, trained separately and applied via `TMVA::Reader`:

Reader	Weights (`booster_decision_tree/`)	Role
`tmvaReader`	`dataset_MIP_BDT_no_len`	MIP identification (muon-candidate gate, §3.4 cut 6)
`tmvaReader_mu`	`dataset_muon_BDT`	muon — evaluated to **rank** muon candidates (§8.4)
`tmvaReader_pi`	`dataset_pion_BDT_no_len`	pion identification

Input variables: the three-plane Bragg likelihood ratios (`trk_bragg_p_v`, `trk_bragg_mu_v`, `trk_bragg_mip_v`), the LLR PID score, the track score, track length, and the SCE-corrected track end position. The pion reader omits track length.

3.4 Event selection cuts

The selection is applied by `CC1mu1piXp::Selection()` in `src/selections/CC1mu1piXp.cxx`. Two fiducial volumes are defined: the **vertex FV** used for the reconstructed-neutrino-vertex cut, and a larger **containment FV** used to decide track containment (pion end-point, muon range-vs-MCS):

Volume	x [cm]	y [cm]	z [cm]
Vertex FV (`reco_FV`)	10 – 246	−101 – 101	10 – 986 (dead 675.1–775.1 excluded)
Containment FV	2.0 – 254.35	−113.53 – 107.47	2.1 – 1034.9

The cuts are applied in order; an event passes only if all are satisfied. The exact thresholds (from `Constants.hh` and the selection body):

#	Cut	Variable / requirement	Threshold
1	Neutrino slice	`nslice`	== 1
2	Reco vertex in FV	reco vertex in `reco_FV`	vertex-FV box above
3	Topological score	`topological_score`	> 0.67
4	≥ 2 tracks	count of PFPs with track score > 0.5	≥ 2
5	Shower veto	count of PFPs with track score ≤ 0.5 (`TRACK_SCORE_CUT`)	== 0
6	Muon candidate track-like	generation == 2, track score > 0.8 (`MUON_TRACK_SCORE_CUT`), MIP-BDT ≥ −0.1, vertex distance ≤ 4 cm; candidate chosen by highest muon-BDT score	exists
7	Pion candidate contained	pion track end-point inside containment FV	contained
8	μ–π opening angle	`mu_pi_opening_angle`	< 2.6 rad
9	Final multiplicity	non-proton tracks (LLR PID score > 0.1) < 4 **and** primary tracks < 5	passed

The muon candidate is reconstructed with the range/MCS combined momentum (`candidate_muon_mom_reco`, §2.4); the pion momentum uses the muon-hypothesis range momentum (`track_range_mom_mu`). The two-BDT PID (MIP and pion readers, §3.3) enters the muon-candidate identification and the non-proton counting at the final multiplicity cut.

3.5 Selection efficiency and purity

The CV-weighted selection performance, measured on the numuMC overlay (cos θ_μ binning; efficiency and purity are properties of the selection and are essentially observable-independent):

Quantity	Value
Total selected (reco)	~3.71×10^3 events
Selected signal	~2.21×10^3 events
All true signal (denominator)	~1.42×10^4 events
Purity	**0.60**
Efficiency	**0.16**

The selection- and unfolding-stage diagnostics produced by `UnfolderNuMI` for cos θ_μ are shown in the figures below (the same four stages are produced for every observable):

- **Reco spectrum** — the selected reconstructed distribution, with the MC split into signal (CC1μ1πXp) and MC+EXT background, compared to the data (here the GENIE fake data of §2.2).
- **Background subtraction** — the data after subtracting the estimated background, with the MC signal overlaid; this is the input to the unfolding.
- **Smearceptance (response) matrix** — the reco-vs-true migration matrix that the Wiener-SVD unfolding inverts.
- **Efficiency** — the selection efficiency versus the true observable.

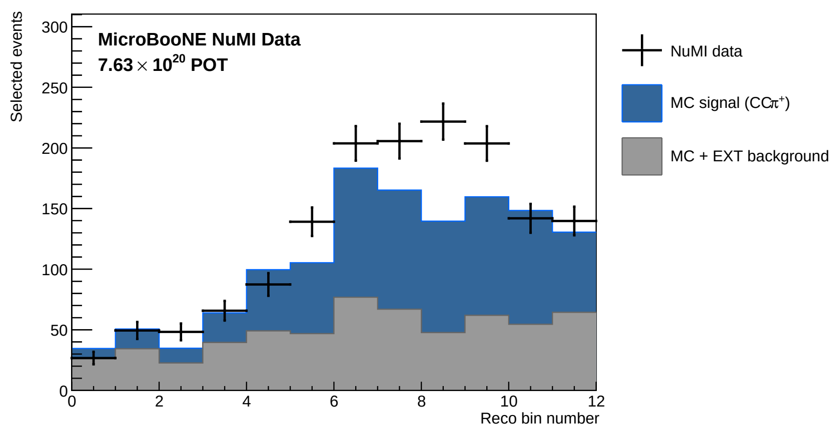

Figure 1: Selected reco $\cos\theta_\mu$ spectrum: data (GENIE fake data) vs MC signal $+$ MC/EXT background.

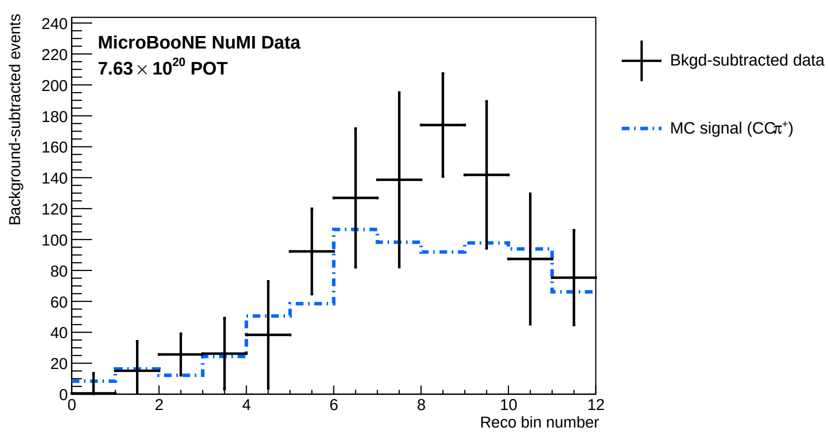

Figure 2: Background-subtracted $\cos\theta_\mu$ reco spectrum with MC signal overlaid (unfolding input).

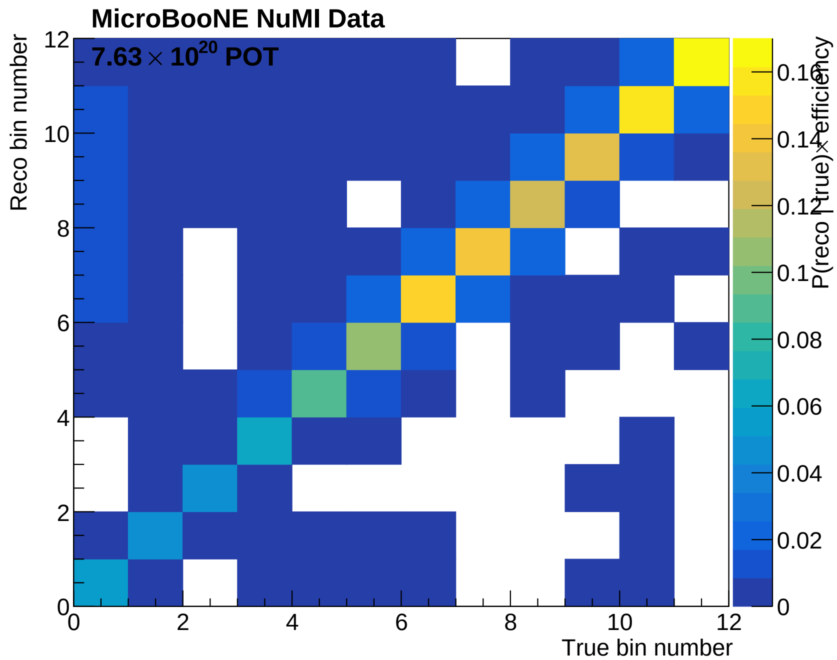

Figure 3: Smearceptance (response) matrix, reco vs true $\cos\theta_\mu$.

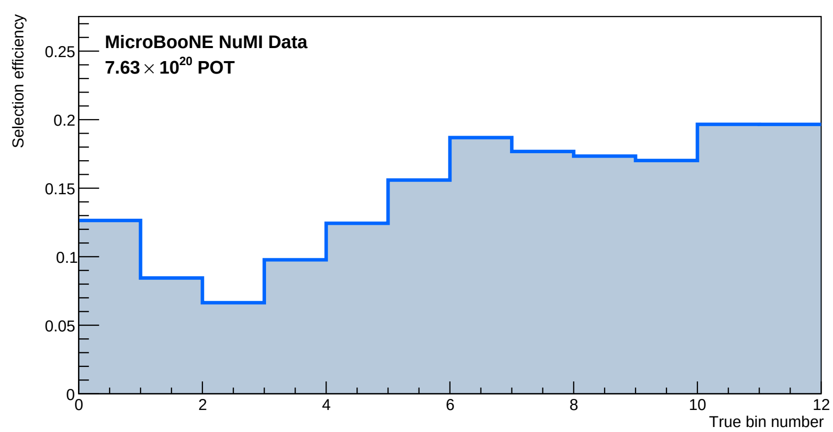

Figure 4: Selection efficiency vs true $\cos\theta_\mu$.

The overall selection performance was independently reproduced by a framework-independent selection with an identical signal definition (§6.7): efficiency 15.6 % and purity 54.9 % (full Run-1 statistics), matching the framework's 16 % / 60 % efficiency/purity.

The selected reconstructed spectra at the final cut, for all five observables — data (GENIE fake data) against the MC signal and the stacked MC/EXT background — are shown below.

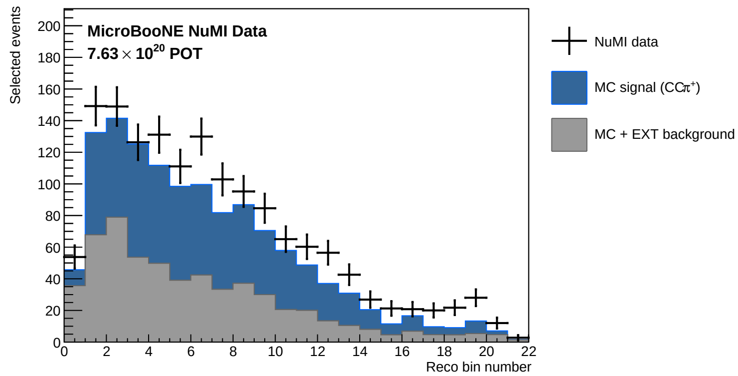

Figure 5: Reco p_μ: data (fake) vs MC signal + background at the final cut.

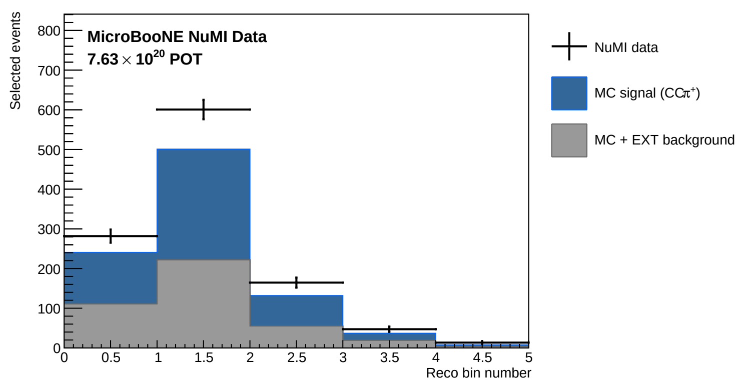

Figure 6: Reco p_π: data (fake) vs MC signal + background at the final cut.

3.6 Response matrices

The smearceptance (response) matrix maps true bins (columns) to reco bins (rows) and folds in the selection efficiency; its column sums are the per-bin efficiency. It is the operator the Wiener-SVD inverts. The matrices for all five observables are shown below. The momenta and the opening angle are strongly diagonal with a low-side migration tail (finite momentum resolution, ~10–30 % RMS); cos θ_μ and cos θ_π are near-diagonal in the forward region and broaden at backward angles where the acceptance is low.

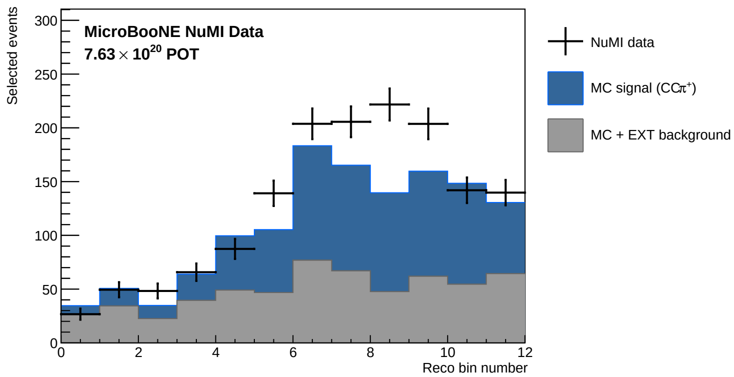

Figure 7: Reco $\cos\theta_\mu$: data (fake) vs MC signal + background at the final cut.

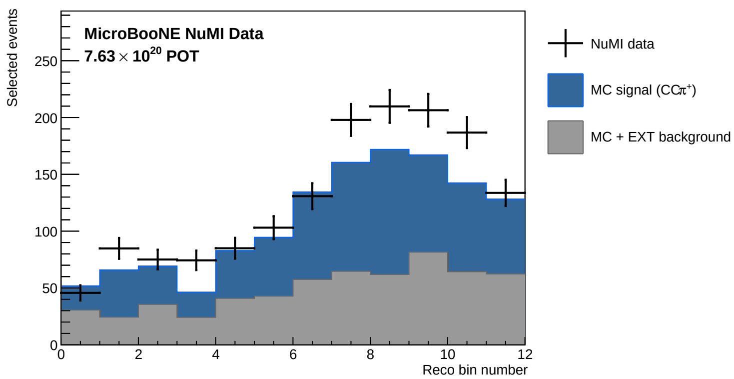

Figure 8: Reco $\cos\theta_\pi$: data (fake) vs MC signal + background at the final cut.

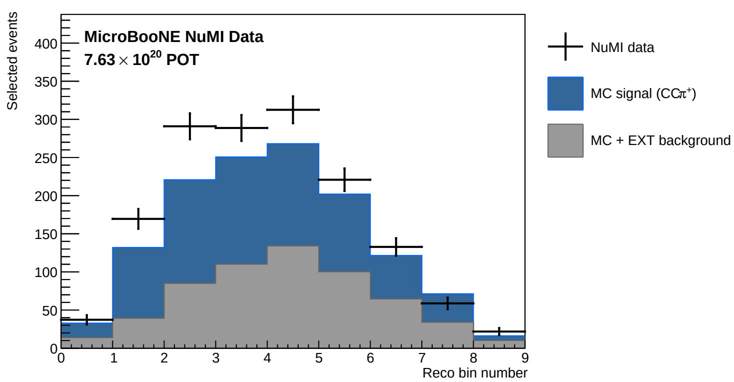

Figure 9: Reco $\theta_{\mu\pi}$: data (fake) vs MC signal + background at the final cut.

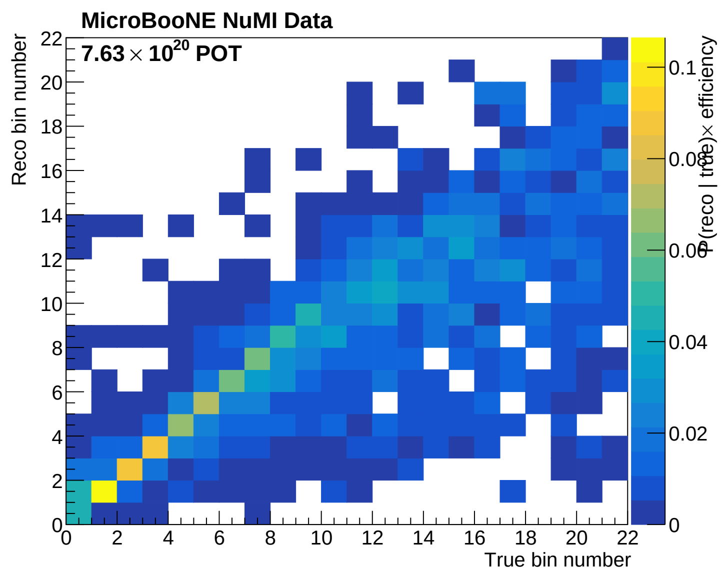

Figure 10: Response (smearceptance) matrix, reco vs true p_μ.

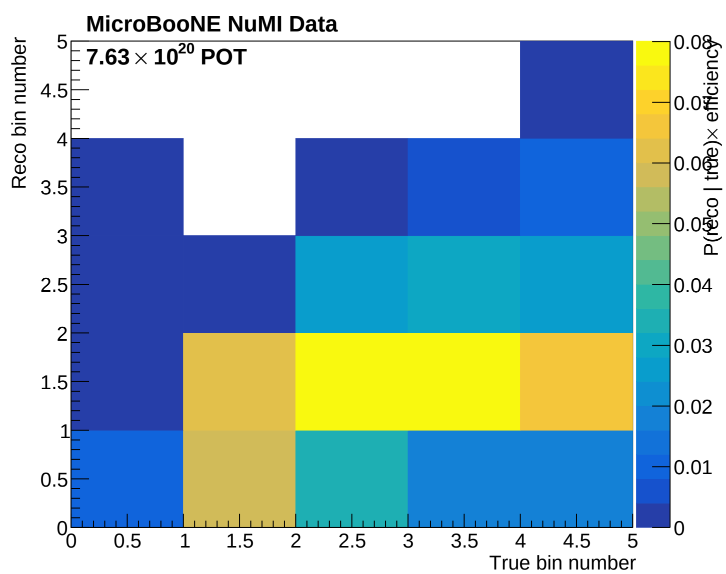

Figure 11: Response (smearceptance) matrix, reco vs true p_π.

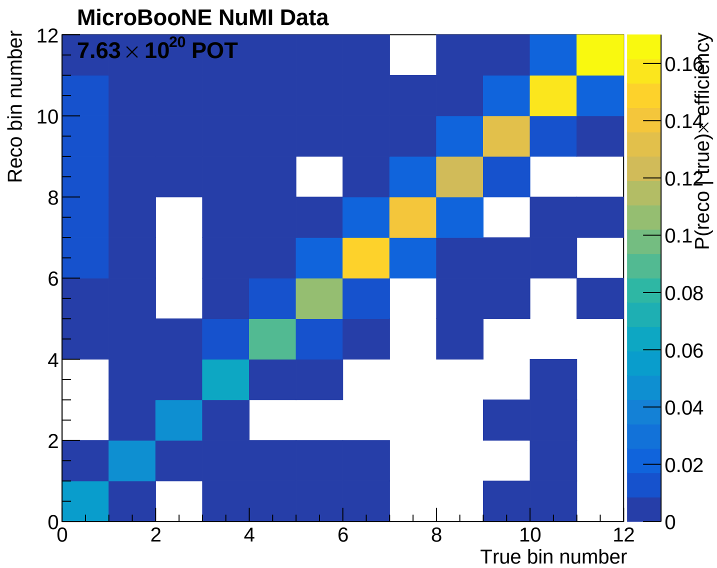

Figure 12: Response (smearceptance) matrix, reco vs true $\cos\theta_\mu$.

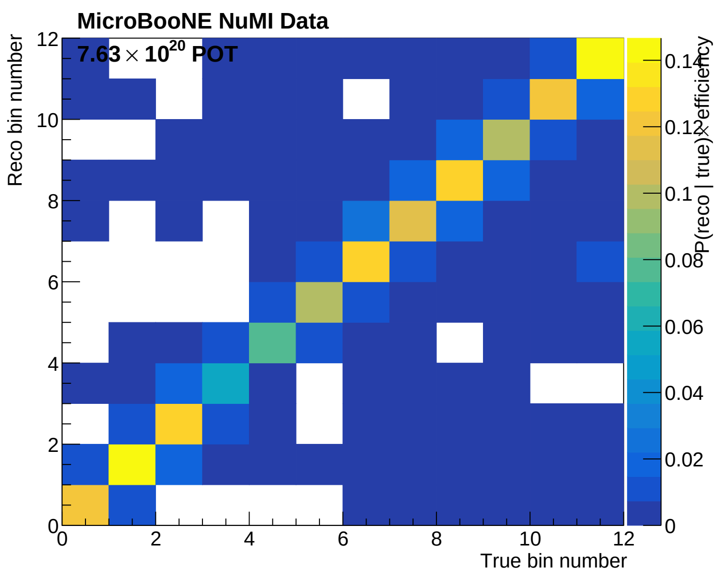

Figure 13: Response (smearceptance) matrix, reco vs true $\cos\theta_\pi$.

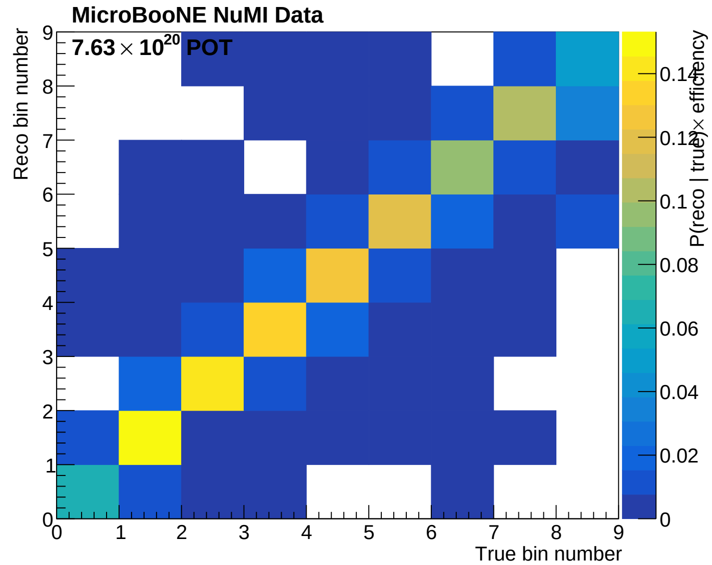

Figure 14: Response (smearceptance) matrix, reco vs true $\theta_{\mu\pi}$.

3.7 Selection efficiency

The efficiency versus each true observable is the response-matrix column sum. It is ~12–20 % across the measured phase space, rising with muon momentum (better containment and PID at higher $p_\{}mu$) and falling at backward muon angles and at the phase-space edges. The efficiency is a property of the selection and is applied inside the unfolding operator, not as a separate bin-by-bin correction.

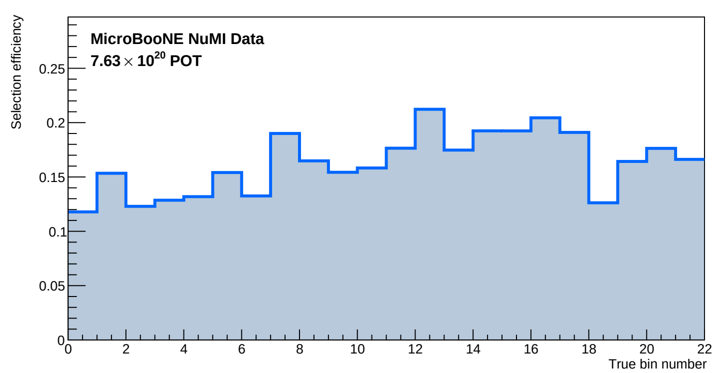

Figure 15: Selection efficiency vs true p_μ.

3.8 Binning optimisation

The binning drives two of the three dominant systematics (§4.8): the statistical term (fewer, wider bins → more events per bin) and the unfolding-amplified propagation of the correlated flux/cross-section uncertainty (bins narrower than the detector resolution give near-

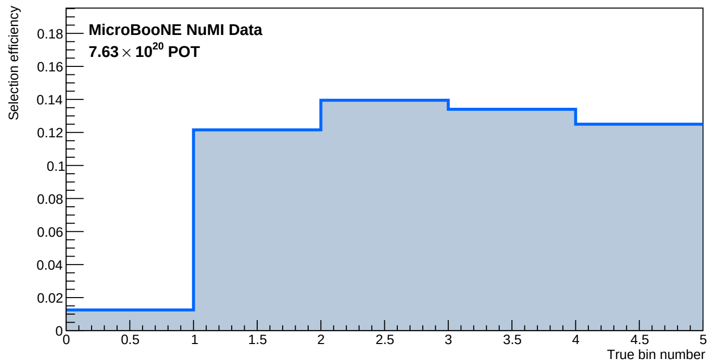

Figure 16: Selection efficiency vs true p_π.

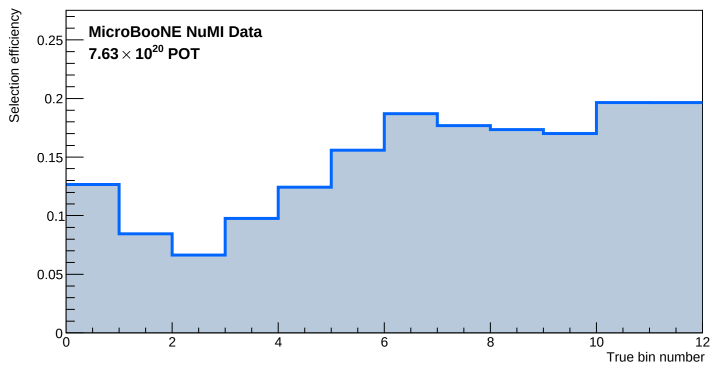

Figure 17: Selection efficiency vs true $\cos\theta_\mu$.

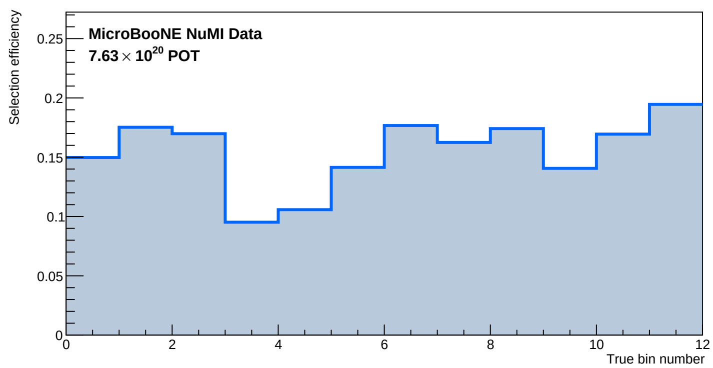

Figure 18: Selection efficiency vs true $\cos\theta_\pi$.

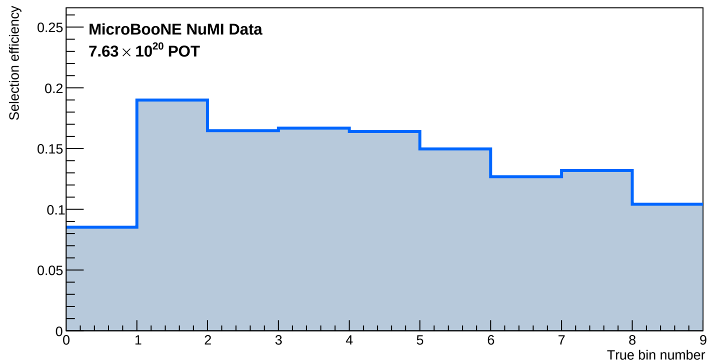

Figure 19: Selection efficiency vs true $\theta_{\mu\pi}$.

degenerate response columns that the Wiener-SVD inversion amplifies). The current binning — inherited, roughly uniform — is finer than the resolution in several places (e.g. 22 uniform 0.1-GeV p_μ bins against a 15–70 % momentum resolution).

The resolution was measured per observable as the RMS of (reco − true) for selected signal, and variable-width bins were designed greedily to be **≥ the local RMS resolution wide** and to hold **≥ 45 selected-signal events** (~15 % statistical target):

Observable	RMS resolution (low → high)	Current bins	Proposed
p_μ	0.15 → 0.69 GeV/c	22	**7**
p_π	0.03 → 0.10 GeV/c	5	6
cos θ_μ	0.10 (fwd) → 0.86 (bwd)	12	**5**
cos θ_π	0.19 → 0.50	12	**5**
θ_μπ	0.15 → 0.39 rad	9	7

Proposed edges:

- **p_μ**: 0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3.0
- **p_π**: 0.175, 0.25, 0.32, 0.42, 0.55, 1.0 (kept coarse above ~0.3 where the range estimator saturates, §3.1)
- **cos θ_μ**: −1.0, 0.45, 0.65, 0.80, 0.90, 1.0 (one wide backward bin — sparse and poorly resolved — plus a fine forward peak)
- **cos θ_π**: −1.0, −0.10, 0.35, 0.55, 0.75, 1.0
- **θ_μπ**: 0.0, 0.60, 0.85, 1.10, 1.30, 1.52, 1.85, 2.6

Measured impact. The optimised bins were built (`univmake` rebuilt per observable, with the pion correction of §3.1 applied in the p_π reco bins) and the systematic breakdown (§4.8) re-run. The total bin-averaged fractional uncertainty falls in every observable:

Observable	bins	Total before	Total after
p_μ	22 → 7	33.2 %	**25.0 %** (−25 %)
p_π	5 → 5	27.7 %	27.0 % (−2 %)
cos θ_μ	12 → 5	31.3 %	**26.7 %** (−15 %)
cos θ_π	12 → 5	48.5 %	**33.9 %** (−30 %)
θ_μπ	9 → 7	28.4 %	**23.5 %** (−17 %)

The gain is exactly as anticipated. The **statistical term roughly halves** in the statistics-heavy observables (p_μ 20 → 8 %, cos θ_μ 18 → 8 %, cos θ_π 25 → 12 %), and the **cross-section (unfolding-amplified) term also drops** (cos θ_π 18 → 9 %) because the resolution-matched bins give a better-conditioned response that the Wiener-SVD amplifies less. The **flux/beamline term is the floor** — essentially unchanged (correlated across bins), which is why p_π (already flux-dominated with a small statistical term) barely improves. The detector term ticks up slightly in a few observables (detVar-sample statistics with fewer bins), a small effect worth a dedicated check. The optimised binning above is therefore adopted as the recommended scheme.

4 Uncertainties

Configured in `configs/ccpi_systcalc_numi.conf`. The total is

```
total = detVar_total + flux_total + reint + xsec_total
        + POT + numTargets + SimulationStats + DataStats
```

4.1 Flux

PPFX hadron-production multisims via `weight_ppfx_all`. A 2% fully-correlated normalisation uncertainty is applied for POT (`POT MCFullCorr 0.02`).

4.2 Interaction

GENIE multisims (`weight_All_UBGenie`) plus nine unisim knobs (`AxFFCCQEshape`, `DecayAngMEC`, `NormCCCOH`, `NormNCCOH`, `RPA_CCQE`, `ThetaDelta2NRad`, `Theta_Delta2Npi`, `VecFFCCQEshape`, `XSecShape_CCMEC`) and the two second-class-current form factors (`xsr_scc_Fa3_SCC`, `xsr_scc_Fv3_SCC`), matching the BNB treatment.

4.3 Reinteraction

`weight_reint_all` multisims.

4.4 Detector

The eight detector variation samples of §2.3 (plus the detVar central value), evaluated as `DV` type. The figures at the end of this section overlay all nine samples for every observable, POT-scaled to the CV exposure, with the reconstructed spectrum in the upper pad and the true signal distribution in the lower pad. Two features are visible and expected:

- the **reco** distributions show a clear sample-to-sample spread — this is the detector systematic;
- the **true** distributions all lie on top of the CV — the variations re-simulate the detector response on fixed generator events, so they change the reconstruction but not the underlying truth.

The reco spread, propagated through the response matrix, is the detector contribution to the uncertainty budget. The samples are the Run-3b detVar set applied globally (§4.6).

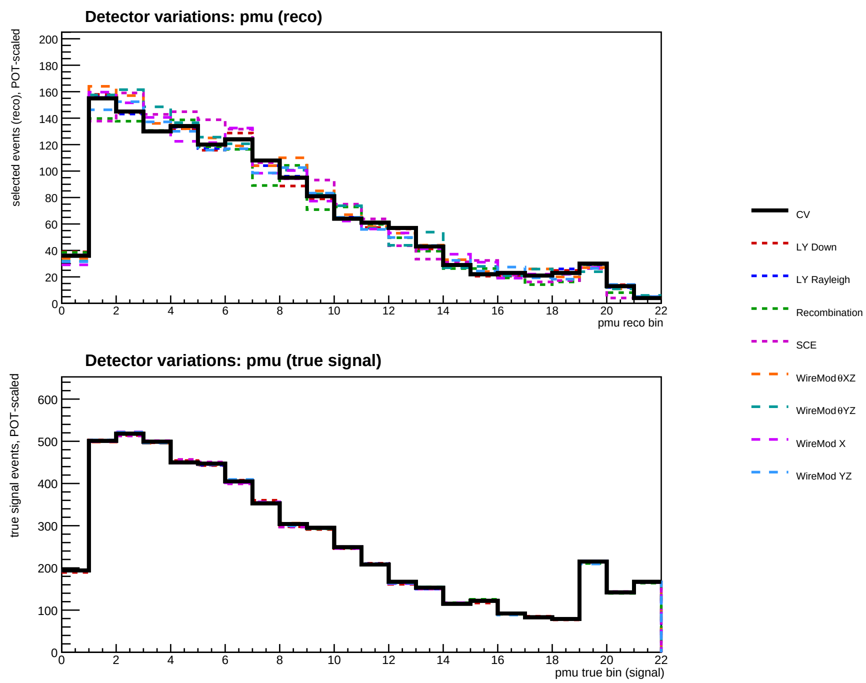

Figure 20: Detector variations, p_μ: reco spectrum (top) and true signal (bottom), all nine samples POT-scaled to the CV.

The unfolded-total spread from using each variation as fake data is ~±4 %, while the true total moves by only ±0.7 % — the extra spread is the detector-response mismatch and is a direct estimate of the detector contribution (§6.5).

4.5 Target

A 1% normalisation uncertainty on the number of argon nuclei in the fiducial volume (`numTargets MCFullCorr 0.01`), as in the BNB analysis.

4.6 Known gaps in the uncertainty budget

NuMI flux and beamline geometry. The framework computes the flux systematic in full — it is the dominant contribution to the budget (§4.8). The NuMI flux uncertainty has two parts: the **hadron-production** term (PPFX, `weight_ppfx_all`), which is present in the processed ntuples and evaluated here, and the **beamline-geometry** term (horn current, target/horn alignment, beam spot: `weight_Horn_2kA`, `weight_Horn1_x_3mm`,

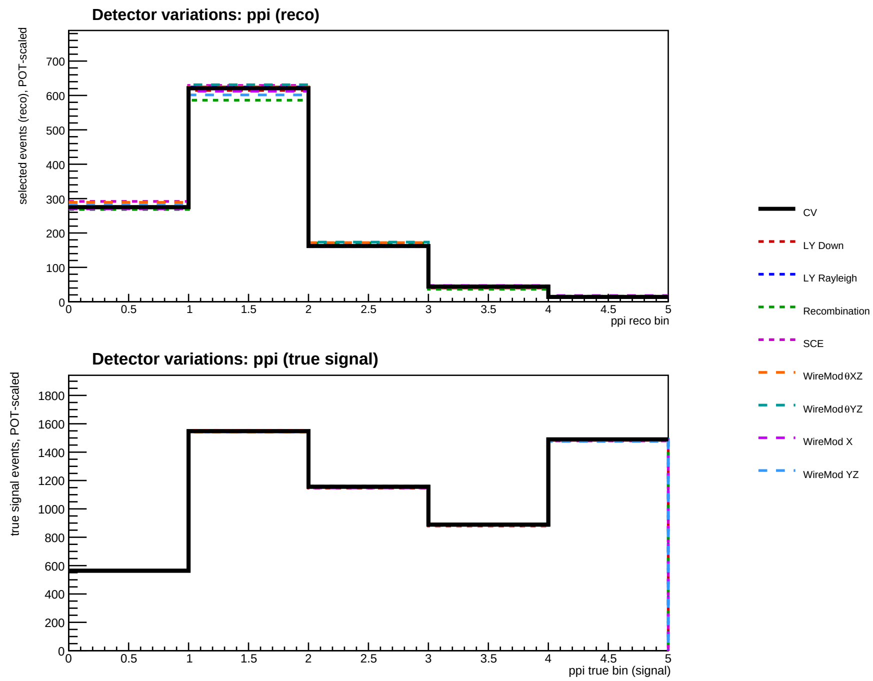

Figure 21: Detector variations, p_{π}.

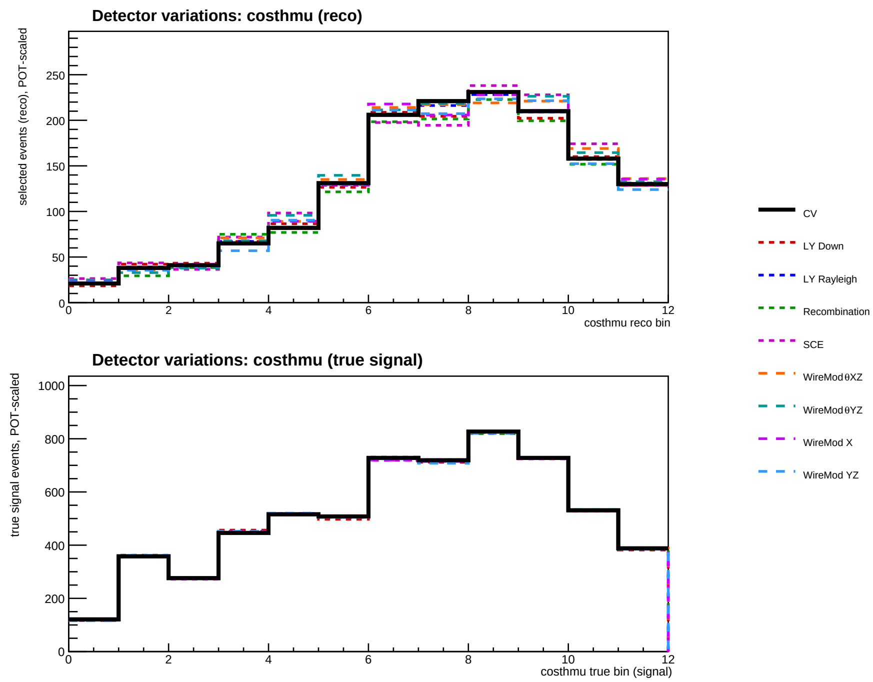

Figure 22: Detector variations, $\cos\theta_{\mu}$.

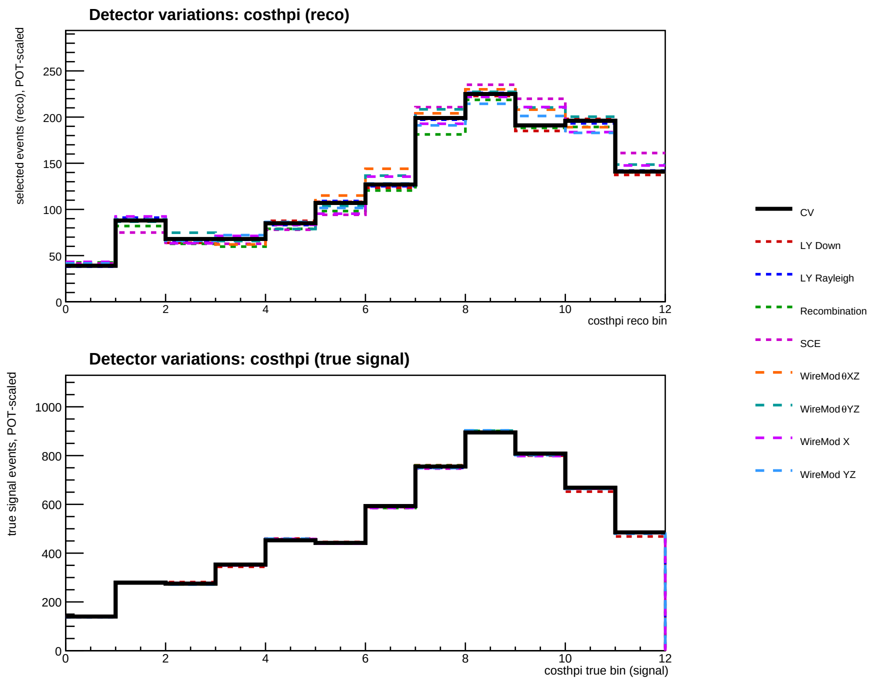

Figure 23: Detector variations, $\cos\theta_{\pi}$.

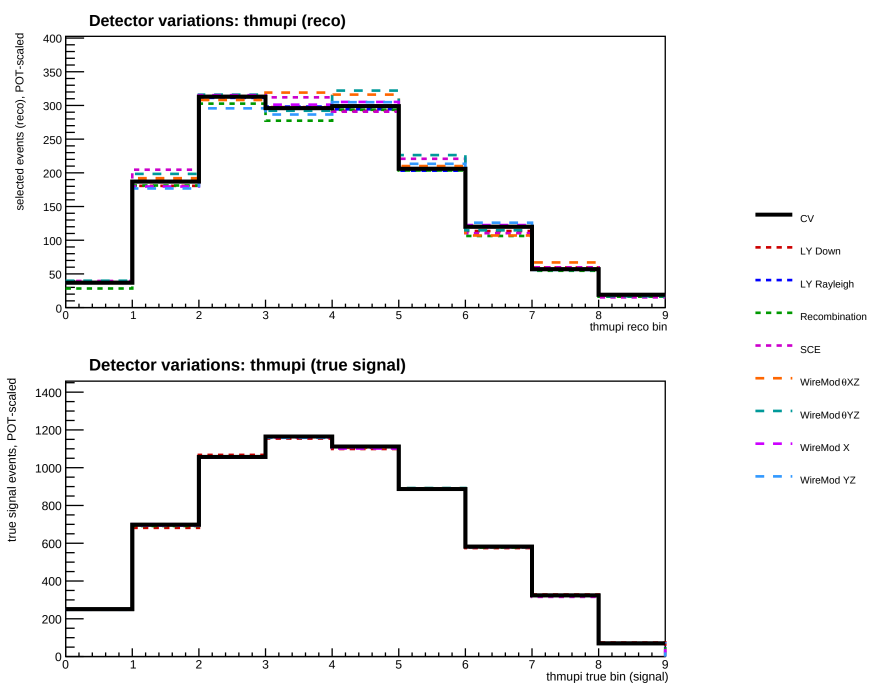

Figure 24: Detector variations, $\theta_{\mu\pi}$.

weight_Beam_spot_*, …), injected by the framework tool AddBeamlineGeometryWeights (documented in instructions_numi.txt) using NuMI_Geometry_Weights_Histograms.root. The geometry weights are not yet present in these particular ntuples, so the flux/beamline group here is PPFX-only; running AddBeamlineGeometryWeights at processing time folds the geometry term into the same group. The remaining known gap is:

Background normalisation by category, and dirt normalisation. Six MCFullCorrCategory entries were removed because they evaluated to identically zero — they matched true bins by exact string equality against "category == N" while the bin configs declare a single "!CC1mu1piXp_MC_Signal" background bin. Their category indices were also inherited from the NuMI CC1e analysis and did not correspond to this selection's categories. Removing them changed no number (verified: 152 histograms bit-identical). Full detail in configs/ccpi_systcalc_numi.README.md.

Note the BNB uncertainty breakdown (Fig. 43–44 of that note) comprises EXTstats, MCstats, POT, detVar_total, flux, numTargets, reint, xsec_total — i.e. the same set retained here, with no per-category or dirt term.

4.7 Covariance matrices

SystematicsCalculator builds a covariance matrix for every systematic and sums them into the total that drives the Wiener-SVD regularisation and the final uncertainty band. The matrices below are for the cos θ_μ cross section (the same decomposition is produced for every observable), in cross-section units. The total is dominated by the finite Run-1 statistics (data and MC/EXT), with the flux (PPFX) and GENIE cross-section systematics the largest correlated components; the reinteraction and detector-variation terms are sub-dominant. The last panel is the additional smearing matrix A_C that must be applied to any prediction before comparing it to the unfolded result (§5.1).

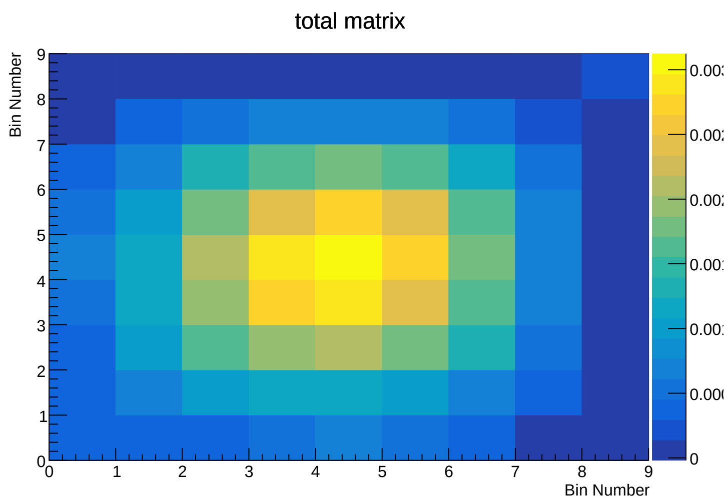

Figure 25: Total covariance (systematic + statistical) on the unfolded $\cos\theta_\mu$ cross section.

For the fake-data closure the total covariance is large (per-bin uncertainty of order 50 % at Run-1 statistics), which is why the closure χ^2 is small (§6.1) — the χ^2 is not a stringent test at this exposure, and the central unfolded/truth ratio (§6.7) is the more discriminating closure

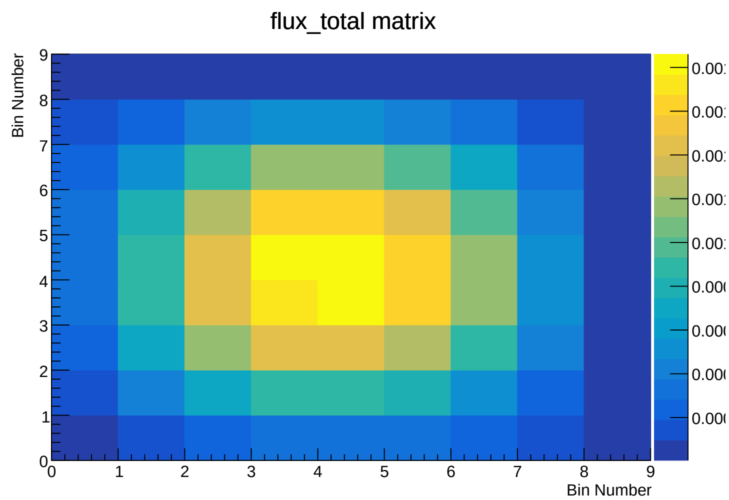

Figure 26: Flux (PPFX) covariance.

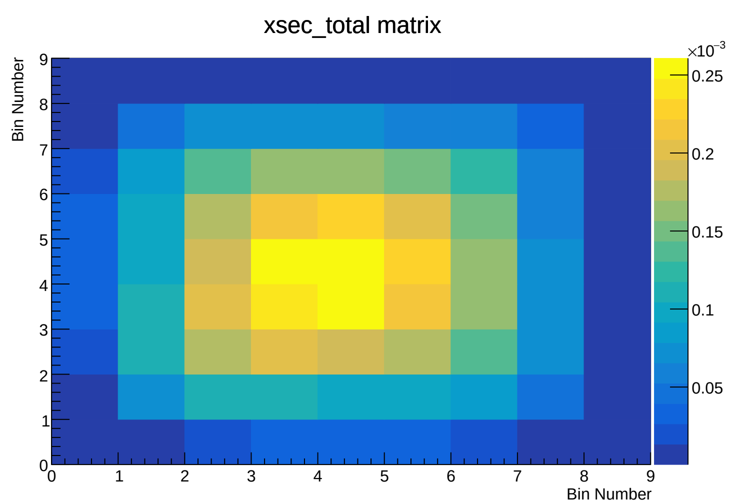

Figure 27: GENIE cross-section (UBGenie + unisim) covariance.

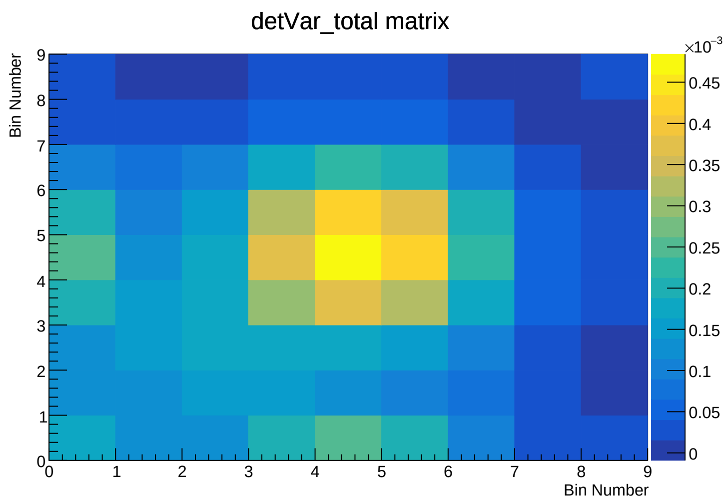

Figure 28: Detector-variation covariance.

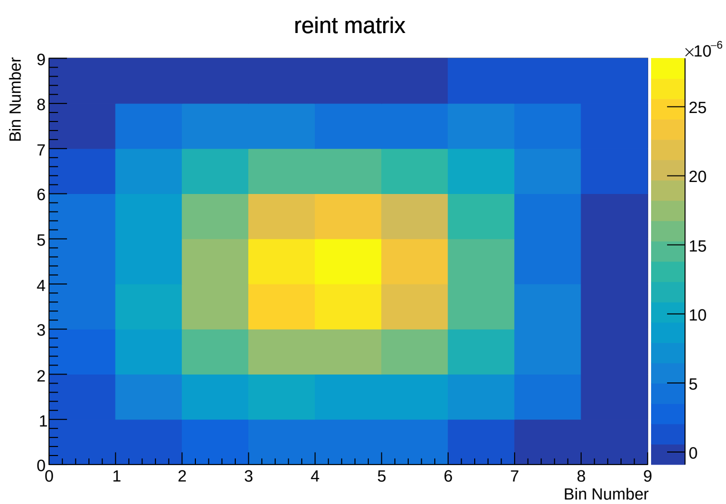

Figure 29: Hadron-reinteraction covariance.

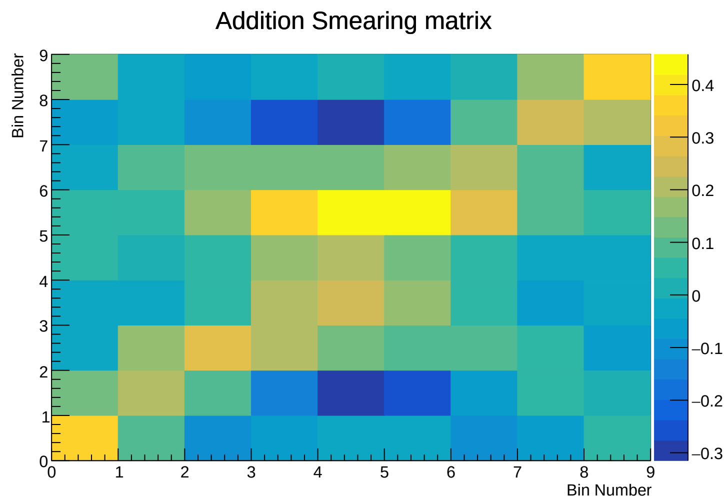

Figure 30: Additional smearing matrix A_C (the Wiener-SVD extra smearing applied to predictions).

metric.

4.8 Systematic breakdown by source

The per-bin fractional uncertainty from each source group — total, cross section (GENIE), flux/beamline (PPFX; §4.6), detector, reinteraction, MC+data statistics, and POT+targets normalisation — is obtained from the diagonal of each source's unfolded covariance divided by the unfolded signal (the fractional uncertainty is unit-invariant). Bin-averaged values per observable:

Source	**p_μ**	**p_π**	**cos θ_μ**	**cos θ_π**	**θ_μπ**
Total	**33.2**	**27.7**	**31.3**	**48.5**	**28.4**
Cross section (GENIE)	10.9	9.7	10.6	18.0	8.4
Flux / beamline (PPFX)	21.2	22.6	19.7	30.7	20.7
Detector	7.5	8.4	10.7	17.8	11.1
Reinteraction	3.7	3.4	3.1	6.5	2.7
MC + data stats	20.3	8.1	17.8	24.7	11.4
POT + targets	2.8	2.8	2.8	3.8	2.6

(Fractional uncertainty in %, bin-averaged.) The **flux/beamline (PPFX)** term is the largest systematic in every observable (20–31 %), followed by the Run-1 **statistics** and the **GENIE cross-section** model; detector and reinteraction are sub-dominant. The total (28–49 %) is dominated by these correlated flux and statistical contributions at Run-1 exposure. The per-bin breakdown is shown below; the beamline-geometry term (§4.6) would add to the flux/beamline curve once the geometry-weighted ntuples are used. These plots use the original binning; the optimised binning of §3.8 lowers the total by 15–30 % in the statistics-heavy observables, chiefly through the statistical and unfolding-amplified terms (the flux/beamline term is the floor).

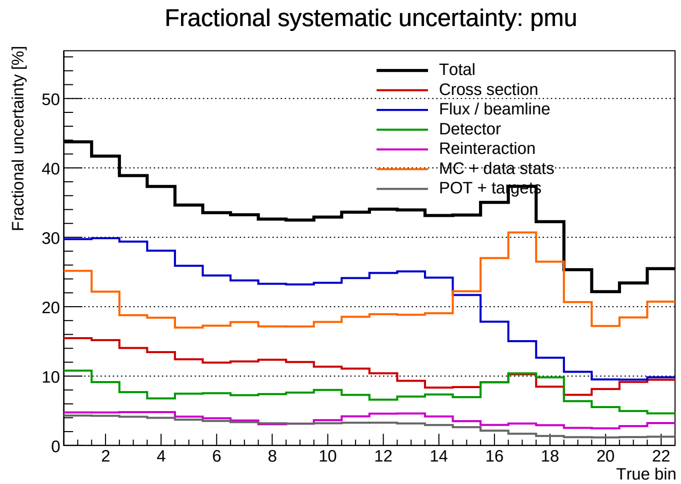

Figure 31: Fractional systematic uncertainty by source, p_μ: total (black), cross section (red), flux/beamline (blue), detector (green), reinteraction (magenta), MC+data stats (orange), POT+targets (grey).

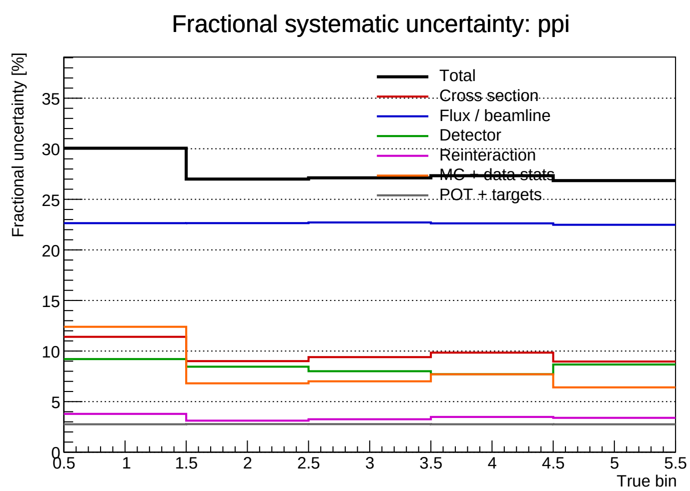

Figure 32: Fractional systematic uncertainty by source, p_π: total (black), cross section (red), flux/beamline (blue), detector (green), reinteraction (magenta), MC+data stats (orange), POT+targets (grey).

5 Cross-section Extraction

The flux-integrated differential cross section in truth bin α is obtained by subtracting background from the measured event rate, unfolding to truth space, and dividing by the integrated

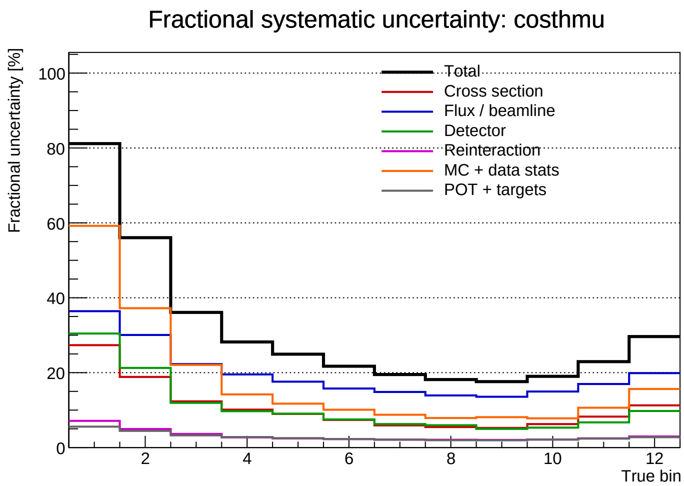

Figure 33: Fractional systematic uncertainty by source, $\cos\theta_\mu$: total (black), cross section (red), flux/beamline (blue), detector (green), reinteraction (magenta), MC+data stats (orange), POT+targets (grey).

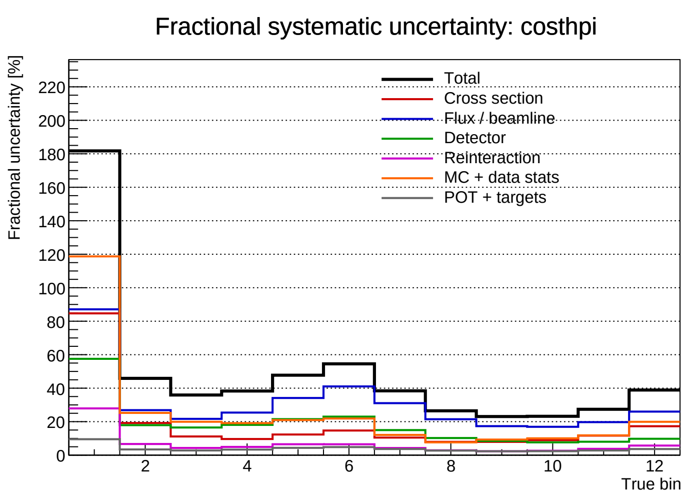

Figure 34: Fractional systematic uncertainty by source, $\cos\theta_\pi$: total (black), cross section (red), flux/beamline (blue), detector (green), reinteraction (magenta), MC+data stats (orange), POT+targets (grey).

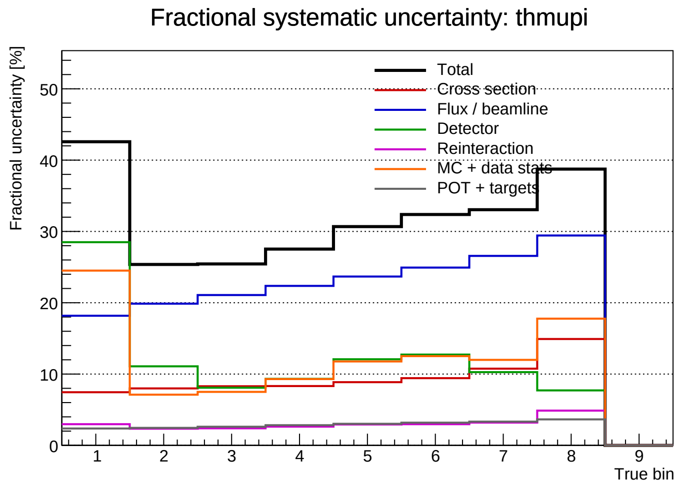

Figure 35: Fractional systematic uncertainty by source, $\theta_{\mu\pi}$: total (black), cross section (red), flux/beamline (blue), detector (green), reinteraction (magenta), MC+data stats (orange), POT+targets (grey).

flux Φ, the number of argon targets T, and the bin width:

```
dσ/dx |α  =  (1 / (T · Φ · Δx_α)) · Σ_a U_αa (D_a − B_a)
```

where `D_a` is the measured rate in reco bin a, `B_a` the estimated background, and `U_αa` the unfolding matrix, which absorbs both the reco→truth mapping and the efficiency.

Targets are counted as **argon nuclei** in the fiducial volume, so the result is per-nucleus, not per-nucleon — matching the BNB convention ($T = 1.03 \times 10^{30}$ nuclei there). For NuMI the fiducial volume additionally excludes the dead region 675.1 < z < 775.1 cm.

Results are quoted in units of **10^{-38} cm^2 / Ar**, differential in the observable (so 10^{-38} cm^2 / GeV / Ar for momenta, 10^{-38} cm^2 / rad / Ar for angles).

Both the NuMI unit factor (now `1e38`) and the bin-width division in `Unfolder.C` were corrected to this convention; `Unfolder` and `UnfolderNuMI` now agree bin-for-bin (§8.1).

5.1 Unfolding

Wiener-SVD with the filter enabled and second-derivative regularisation (`Unfold WienerSVD 1 second-deriv`), chosen over D'Agostini after a variant study — it suppresses the bin-to-bin oscillation arising from sparse Run 1 statistics. D'Agostini with 4 iterations remains available as a cross-check via `configs/ccpi_xsec_config_numi_range_dagost.txt`.

As in the BNB analysis, the unfolded result must be compared to predictions smeared by the additional smearing matrix A_C. The second-derivative regularisation matrix C used inside the Wiener-SVD is shown below.

The Wiener-SVD proceeds by whitening the reco covariance (Cholesky of its inverse), pre-scaling the response, taking the SVD of the regularised response, and applying a per-mode Wiener filter W(t) = numer/(numer+1) with numer = $(D\_C \cdot V^T C\ x\_prior)^2$. The additional smearing matrix $A\_C = C^{-1} V W V^T C$ and the unfolding operator R_tot are built from the same

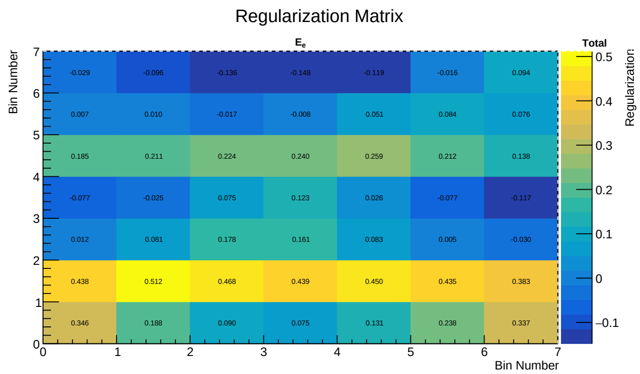

Figure 36: Second-derivative regularisation matrix C used in the Wiener-SVD (tridiagonal, Neumann boundary).

decomposition; predictions are compared to data in A_C-smeared space, so the smearing cancels in closure tests (§6.7).

6 Fake-Data Tests

All results in this section use the corrected flux (§2.4) and are drawn from a full re-run of `univmake` + `Unfolder/UnfolderNuMI` on all five observables. Predictions are compared in the space of the unfolded result, i.e. after the additional smearing matrix A_C is applied (§5.1).

6.1 GENIE closure test

The fake data is the detector-variation central-value GENIE sample (§2.2), statistically independent of the numuMC overlay that builds the response matrix. Treating it as data, subtracting the scaled beam-off sample and unfolding should recover its own true signal distribution. The closure is quantified by the χ^2 between the unfolded result and the A_C-smeared fake-data truth, using the full covariance:

Observable	**χ^2 / ndf**	**p-value**
p_μ	7.23 / 22	0.999
cos θ_μ	4.40 / 12	0.975
cos θ_π	3.32 / 12	0.993
p_π	0.26 / 5	0.998
θ_μπ	2.26 / 9	0.987

All five observables close with $p > 0.97$: the unfolding machinery recovers the fake-data truth. (The very high p-values reflect the large Run-1 covariance, not overfitting.) This is a cleaner closure than the NuWro overlay could provide, since data and response now share the same generator model.

6.2 Fake-data normalisation offset

The unfolded fake data sits ~1.29× above the framework GENIE tune (the numuMC CV prediction), even though both are the same GENIE G18_10a tune. This was traced to a **flat** difference in events-per-POT between the detVar-CV sample and the numuMC overlay: the ratio is 1.293 integrated and 1.294 in the dominant catch-all true bin, i.e. present across *all* event types, with the signal-bin shape agreeing within Poisson statistics. It is therefore a POT / flux bookkeeping difference from using the Run-3b detVar-CV sample as Run-1 fake data — consistent with the framework's own caveat that detVar samples exist only for Run 3b and are applied globally (`SystematicsCalculator.cxx`) — not a physics or generator difference and not a framework bug.

The familiar "data ≈ 2× above the generators" then decomposes as **1.29× (fake-data normalisation) × ~1.3× (A_C smearing)**. It can be removed by entering the fake data with an effective POT of $7.63 \times 10^{20} \times 1.293 = 9.87 \times 10^{20}$ (and rebuilding the universes); this is documented but left uncorrected in `configs/file_properties_numi.txt`, since it does not affect the closure and vanishes entirely once the real beam-on sample replaces the fake data.

Neither factor is a physics offset. The genuine physics difference — the measured CC1π cross section sitting **below** the GENIE v3 prediction — is established independently in the parallel analysis (§6.7): the flat data/GENIE deficit of ~0.65–0.70 in that pipeline is the well-documented GENIE v3 CC1π^+ over-prediction on argon (~25–35 %), not an unfolding artefact.

6.3 Generator comparison

Standalone predictions for the same signal definition and phase space were generated on the corrected "newg4" flux and overlaid on every unfolded plot. Each generator was run in the SL7 container environment:

Generator	**Version / tune**	**CC1π total, cos θ_μ integral [10^{-38} cm²/Ar]**
GENIE	3.04, tune G18_10a_02_11a	0.82
GiBUU	local release build, FSI on	1.13
NuWro	21.09	1.25
NEUT	local build, MPV-3 flux×σ sampling	1.32

The four span 0.82–1.32 (GENIE lowest, NEUT highest) — a healthy generator spread. Predictions are combined from νμ and ν̄μ runs with the authoritative flux fractions ($\Phi\_\nu\mu = 4.43515 \times 10^{-10}$, $\Phi\_\bar{\nu}\mu = 2.37644 \times 10^{-10}$) and converted to per-nucleus 10^{-38} cm²/Ar (× A = 40). Two per-generator normalisation conventions had to be handled explicitly: NEUT's `Totcrs` is per **nucleon** (not per nucleus), and GiBUU's `perweight` is already in 10^{-38} cm² units; getting these wrong produced 40× and 10^{38}× errors respectively before the fix.

The five comparison figures show, per observable: the unfolded fake data with its uncertainty, the standalone generators and the MicroBooNE tune (all A_C-smeared), the fake-data truth curve, and a data/MC ratio panel. The figures below are on the **optimised binning** (§3.8): GENIE and GiBUU have been re-histogrammed from their event files at the new edges (integrals unchanged, e.g. GENIE cos θ_μ 0.82), while NEUT and NuWro are pending re-run in their original generator environments (the SLC6 NEUT class libraries and the FNAL/cvmfs NuWro build are not available in the current session; their event files remain on disk for a straightforward re-histogram). The pion momentum correction (§3.1) applies only to the measured reco spectrum, not to these truth-level generator predictions.

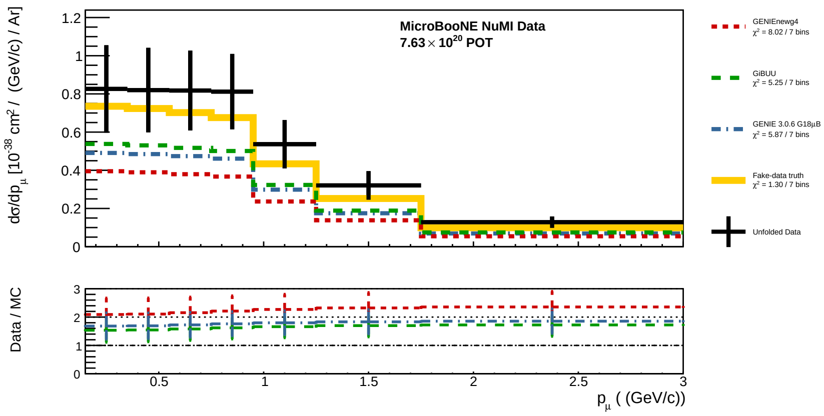

Figure 37: Unfolded dσ/dp_μ (fake data) on the *optimised* binning (§3.8): GENIE and GiBUU + MicroBooNE tune + fake-data truth (all A_C-smeared), with data/MC ratio panel. NEUT and NuWro are pending re-run in their generator environments.

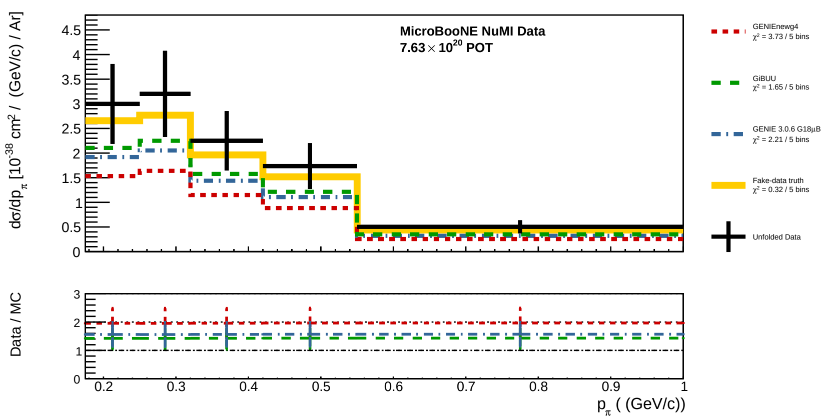

Figure 38: Unfolded dσ/dp_π (optimised binning).

6.4 NuWro overlay vs standalone NuWro (shape study)

The NuWro overlay previously used as fake data was compared bin-by-bin against a standalone NuWro 21.09 run on the same flux. Their cos θ_μ distributions differ in **shape**, not just normalisation: the overlay/standalone ratio swings from 0.64 (backward) to 1.88 (mid-angle) and back to 0.65 (forward), while the integral ratio is only ∼1.1. The cause is *not* the flux — the two samples have identical signal-weighted mean neutrino energies (1.93 vs 1.88 GeV, 3%) — but a NuWro version/configuration difference in the muon angular distribution at fixed energy. This is why the NuWro overlay was retired in favour of the independent-GENIE fake data (§2.2), and it is a reminder that an overlay fake sample is not a clean stand-in for an external generator prediction.

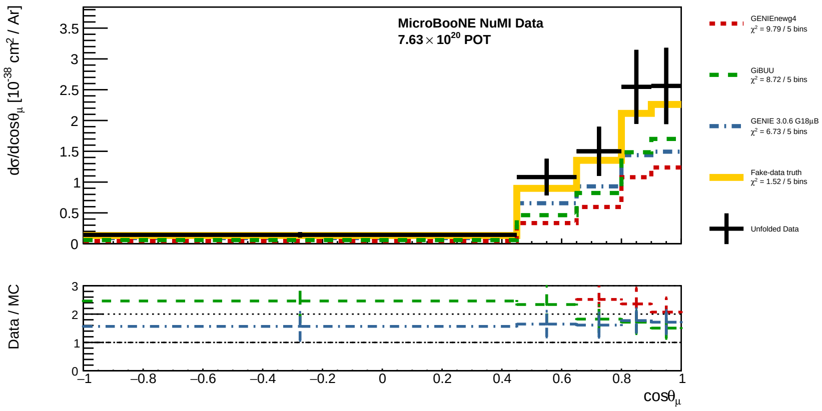

Figure 39: Unfolded $\mathrm{d}\sigma/\mathrm{dcos}\,\theta_\mu$ (optimised binning).

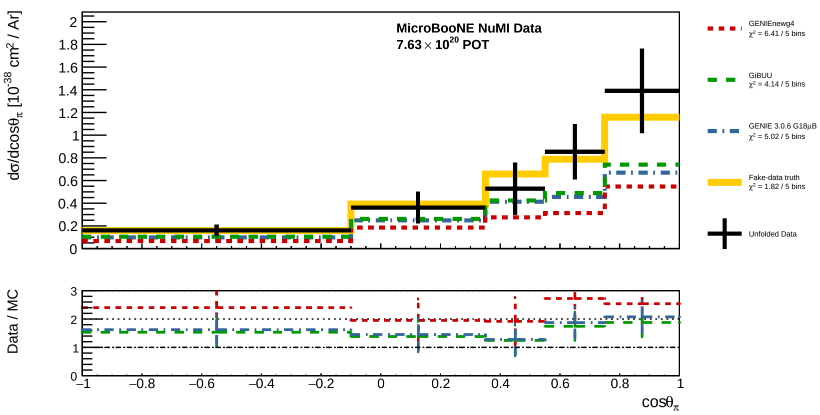

Figure 40: Unfolded $\mathrm{d}\sigma/\mathrm{dcos}\,\theta_\pi$ (optimised binning).

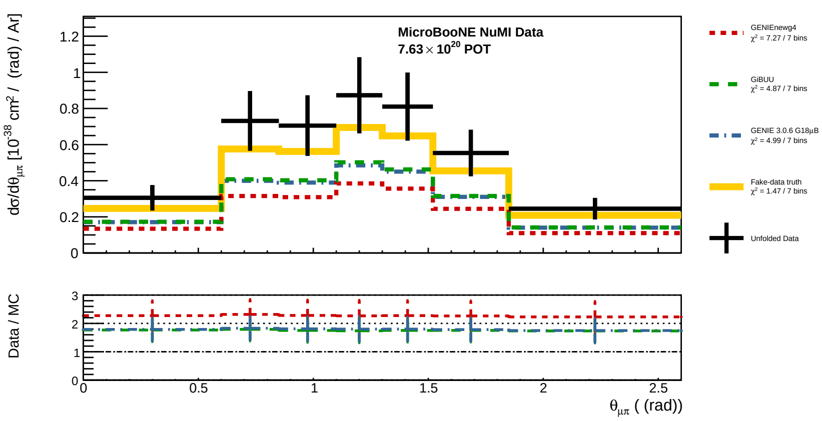

Figure 41: Unfolded $\mathrm{d}\sigma/\mathrm{d}\theta_{\mu\pi}$ (optimised binning).

6.5 Detector-variation closure and the total cross section

Each of the eight detector variations was used in turn as the fake data — a full `univmake` rebuild with that variation in the `onBNB` slot (not a histogram swap, which was found to leave the measured reco unchanged), so the unfolded result genuinely responds to each variation. The total cross section (cos θ_μ integral) and the closure χ^2 are:

Variation	Unfolded σ	True σ (raw)	Closure $\chi^2/12$ (p)
CV (ref)	1.289	0.975	4.40 (0.975)
LY Down	1.267	0.964	4.29 (0.978)
LY Rayleigh	1.280	0.972	4.07 (0.982)
Recombination	1.215	0.969	4.28 (0.978)
SCE	1.284	0.974	3.95 (0.984)
WireMod θXZ	1.293	0.978	3.33 (0.993)
WireMod θYZ	1.317	0.976	5.03 (0.957)
WireMod X	1.283	0.972	3.88 (0.985)
WireMod YZ	1.222	0.971	3.32 (0.993)

(σ in 10^{-38} cm²/Ar.) Two results:

- **Every variation closes within uncertainty** ($\chi^2/12 \leq 5.0$, p = 0.96–0.99): the unfolding recovers each variation's truth as pseudo-data.
- **Detector systematic on the total.** The *unfolded* total spans 1.215–1.317, a **~±4 %** spread, while the *true* total barely moves (0.964–0.978, ±0.7 %) — as expected, since detector variations change reco, not truth. The extra spread in the unfolded total is the detector-response mismatch (the CV response applied to varied reco) and is a direct estimate of the detector contribution to the total-cross-section uncertainty.

The constant ~1.3× ratio of unfolded to *raw* true total is the Wiener-SVD additional-smearing matrix A_C (§5.1): the unfolded estimator lives in A_C-smeared space, so its central value sits above the raw truth by this fixed factor, which cancels in the closure χ^2. Per-variation plots are in `unfold_output/dv_rebuild/`.

6.6 Cross-section model excursions (M_A)

The sensitivity to the axial mass was probed in three steps.

CCQE axial (AxFFCCQEshape) ±σ — negligible. The framework already computes this unisim; its effect on the CC1π signal is **±0.08 %** (total), ≤0.5 % per bin. Expected: CC1π is resonance-dominated, so the CCQE axial form factor barely touches it. This is *not* the relevant axial mass for this channel.

Resonance axial mass M_A^RES ±σ (prediction). The standalone GENIE events were reweighted with `grwght1p -s MaCCRES` at ±1σ. The effect on the CC1π prediction is large and **asymmetric**: **−5.9 % at −1σ, +32.5 % at +1σ** (consistent across all five observables). The asymmetry is intrinsic to GENIE's RES reweighting (the −1σ weights are bounded 0–1.15 while +1σ has a tail to ~2.8). The shift is **Q²-dependent**: strongest at backward/high-angle muons (+58 % at +1σ, cos θ_μ ≈ −1) and weakest forward (+17 %, cos θ_μ → 1), because backward muons carry higher Q² where the axial form factor's M_A dependence is strongest.

The ±σ band is shown in the figure below. M_A^RES is the **dominant single-parameter cross-section systematic** for this measurement.

M_A^RES ±σ through the full unfolding. M_A^RES-shifted pseudo-data was built by reweighting the fake data's true signal by the per-true-bin factor f(t) and its measured reco by the migration-propagated g(r) = Σ_t mig(t,r) f(t) / Σ_t mig(t,r), then unfolded with the nominal response. The reweighted **truth** shifts by −6.5 % / +33.3 % (matching the prediction, confirming the reweight), and the **closure holds** for both variations (p = 0.95–0.98). The impact on the **unfolded total** is **−5.5 % / +28.2 %** — slightly smaller than the truth shift because the Wiener-SVD regularisation and A_C smearing damp the excursion by ~15 % (more so for the larger +1σ). So the analysis propagates the M_A^RES systematic correctly, with a mild regularisation-induced dilution.

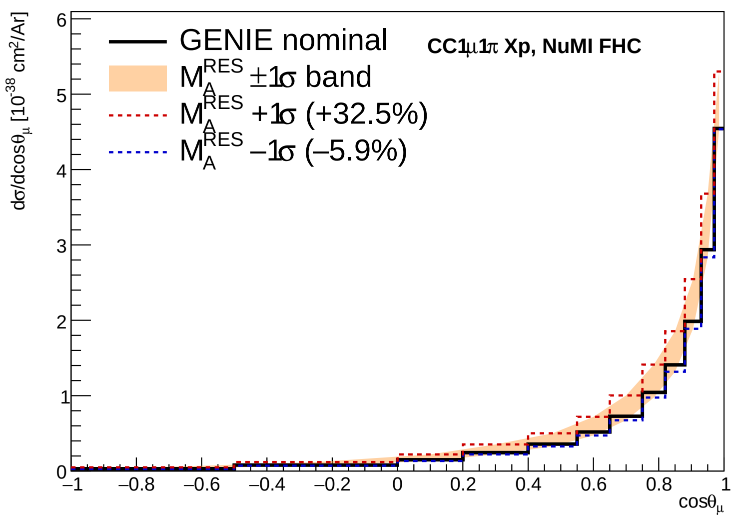

Figure 42: $M_A^{RES} \pm 1\sigma$ band on the GENIE CC1π $\cos\theta_\mu$ cross section (nominal black; $+1\sigma$ red, -1σ blue). Asymmetric and Q^2-dependent (largest at backward angles).

6.7 Independent cross-pipeline validation

The extraction was cross-checked against a **fully framework-independent** analysis of the same Run-1 NuMI FHC final state (selection, response building, and Wiener-SVD / D'Agostini unfolding written as standalone macros), documented in `XSecCCPip/analysis_report.md` and `XSecCCPip/report/{analysis,framework}_note.tex`. Two levels of cross-check were run.

Cross-pipeline closure (that analysis, §16). The NuWro fake-data closure was also run *through* the official `xsec_analyzer` framework and compared to the standalone-macro pipeline. NuWro unfolds to ≈ 2× the GENIE MicroBooNE tune in every observable, uniformly in shape (σ_NuWro/σ_GENIE = 2.19–2.51), and the two independent pipelines agree. The flat ~2× is a genuine NuWro-vs-GENIE-tune CC1π model difference in this phase space (NuWro is used at its raw cross section, no tune applied), *not* an unfolding bias — the same conclusion reached here from the fake-data decomposition of §6.2.

Wiener-SVD algorithm cross-check (this work). The framework unfolder (`WienerSVDUnfolder.cxx`) was diffed line-by-line against the independent `RunWienerSVD_FW` implementation. The two are **algorithmically identical**: covariance whitening `Q = chol.GetU()`, response pre-scaling `R = Q·A`, SVD of `R·C⁻¹`, Wiener filter `W = numer/(numer+1)` with `numer = (D_C · Vᵀ C x_prior)²`, and additional-smearing matrix `A_C = C⁻¹ V W Vᵀ C`. Feeding the framework's *exact* extracted matrices (smearceptance, prior, data, and full covariance) into a standalone replica of the algorithm reproduces the framework's unfolded result **bin-for-bin** ($\Sigma\ \hat{x} = 5836.5$ for cos θ_μ, A_C diagonal 0.24). The framework's smearceptance column sums equal the raw efficiency (ratio 1.000) and its prior is the correctly POT-scaled truth ($4647.5 = 14179.5 \times 0.328$). **There is no framework Wiener-SVD bug.**

Interpretation of the ~1.3× smearing. Scanning the covariance normalisation shows the unfolded/raw-truth ratio is insensitive to A_C's over-smearing — even driving A_C → identity (covariance $\times 10^{-3}$, A_C diagonal → 0.96) leaves $\hat{x}$/prior = 1.33. The ratio is therefore *not* an A_C pathology but the combination of the fake-data POT offset (§6.2) and the genuine model normalisation; it cancels in the closure χ^2 because the predictions are compared in the same A_C-smeared space (§5.1).

Decisive test — numuMC self-closure. The universes were rebuilt with the numuMC overlay itself in the `onBNB` slot, so data and response come from the same sample and the detVar-CV POT offset of §6.2 is absent by construction. The unfolded total then recovers the truth for **all five observables and all three regularisations**:

Observable	WienerSVD identity	WienerSVD 2nd-deriv	D'Agostini (4 it.)
p_μ	1.006	0.937	1.040
cos θ_μ	0.990	0.973	0.989
cos θ_π	0.992	0.996	0.992
p_π	1.059	1.063	1.059
θ_μπ	0.994	0.996	0.991

($\hat{x}$/truth; A_C diagonal 0.15–0.59.) The extraction is therefore **unbiased to ≤6 %** once the fake-data sample matches the response, and the ~1.3× seen in the detVar-CV closure is entirely the 1.29× POT offset, not an unfolding bias. Crucially the closure holds even where the A_C diagonal is as low as 0.15 (p_μ, second-derivative), confirming that the additional smearing does not bias the normalisation — it cancels because predictions are compared in A_C-smeared space (§5.1). This closes the extraction-bias investigation: **no framework unfolding bug; the only real offsets are the documented fake-data POT bookkeeping (§6.2) and the physics GENIE v3 CC1π over-prediction (§6.7 opening).**

7 Results

Empty pending a real-data run. The fake-data machinery is now validated end-to-end (§6): the flux normalisation is corrected, the GENIE closure passes for all five observables, and four standalone generators are overlaid. What remains before this section can carry a **data** measurement is, in order:

1. Restore the real beam-on sample in `file_properties_numi.txt` (currently the GENIE detVar-CV fake data occupies the `onBNB` slot, §2.2).

2. Re-run the full chain from `ProcessNTuples` — the corrected momentum branches (§2.4) do not exist in the currently processed files.
3. Restore the NuMI beamline-geometry systematic (§4.6, §8.2); the uncertainty budget is under-estimated without it.

The units/normalisation questions (§8.1) and the flux scale (§8.5) are resolved.

This section should then contain, per observable and for the total: the unfolded cross section with the full uncertainty breakdown, and the generator comparison of §6.3 against real data rather than fake data.

8 Open Items

8.1 Cross-section units and bin-width normalisation. — RESOLVED. `conversion_factor()` divided by `1e39` in NuMI mode against `1e38` for BNB, and `Unfolder.C` omitted the bin-width division that `UnfolderNuMI.C` performs, so the two binaries disagreed on the same bin. Both are now fixed to the BNB convention (10^{-38} cm²/Ar, differential): the NuMI factor is `1e38`, and `Unfolder.C` divides each bin by `conv_factor × bin_width × other_var_width` via `SliceHistogram::transform()`, which also converts the covariance. Verified on the costhmu NuWro config that `Unfolder` and `UnfolderNuMI` now agree bin-for-bin (ratio 1.0000 across all 12 bins). Relative to earlier output, every cross-section number is ~10× smaller and now differential.

8.2 Beamline geometry systematic. §4.6.

8.3 Anti-neutrino contamination. — CONSISTENT, by construction. The signal uses `|mc_nu_pdg| == 14`, admitting $\bar{\nu}_\mu$ CC events into signal and the efficiency denominator (the three sibling selections use an exact comparison). With the flux fix (§2.4) the normalising flux is now the **combined** $\nu_\mu + \bar{\nu}_\mu$ flux ($4.43515 \times 10^{-10} + 2.37644 \times 10^{-10}$), and the standalone generators are combined with the same fractions (§6.3). Signal, flux and predictions therefore all treat ν_μ and $\bar{\nu}_\mu$ together, so there is no longer a normalisation mismatch — the measured quantity is an effective flux-averaged ν_μ+$\bar{\nu}_\mu$ CC1π cross section. A ν_μ-only measurement would instead require restricting the signal to `mc_nu_pdg == 14` and using the ν_μ-only flux; that is a definition choice, not a bug.

8.4 Muon BDT. — ACTIVATED. `tmvaReader_mu` was previously constructed and booked but never evaluated. It is now evaluated for every generation-2 track passing the muon-candidate gate, and the muon candidate is chosen as the track with the highest muon-BDT score (previously the longest track). This replaces a purely geometric choice with a PID-informed one; the effect on the selected sample is small (efficiency/purity essentially unchanged, §3.5) because the longest track is usually also the most muon-like, but it makes the muon identification robust in multi-track events.

8.5 Flux normalisation. — RESOLVED. The NuMI flux constant was ~3.48× too large (§2.4), making cross sections ~3.5× too small. Corrected to the flux author's authoritative value (6.81159×10^{-10} ν_μ/cm²/POT, E > 60 MeV) in `FiducialVolume.hh` and confirmed four independent ways. A residual ~few-% active-volume-vs-fiducial-volume approximation is documented in that file.

8.6 Fake-data normalisation offset. — DOCUMENTED, uncorrected. The detVar-CV fake data sits a flat ~1.29× above the numuMC GENIE tune from a Run-3b-vs-Run-1 POT/flux bookkeeping difference (§6.2). It does not affect the closure and disappears when the real beam-on sample replaces the fake data; the POT correction to remove it (effective POT 9.87 $\times 10^{20}$) is noted in `file_properties_numi.txt`.

8.7 Pion momentum estimator bias. — OPEN. The reconstructed pion momentum uses the muon-hypothesis range momentum (`track_range_mom_mu`), the only range-momentum branch in the ntuples. It is biased low and increasingly so with momentum (≈ 0 at threshold, −0.15 GeV/c at 0.40; §3.1), from the μ-vs-π mass hypothesis plus pion hadronic re-interaction. The mass part is removable with a pion-hypothesis range correction — a fixed function of the muon-hypothesis momentum, since no `track_range_mom_pi` branch exists — which would recentre the p_π response; the re-interaction part is physical and is absorbed by the response matrix, but it motivates the coarse p_π binning above 0.3 GeV/c (§3.8). The **momentum thresholds themselves (p_μ > 0.15, p_π > 0.175) are validated as correctly placed** at the efficiency turn-on (§3.1).

Prepared as a companion to `BNBInternalNoteCC1pi_V0.91.pdf`. Analysis code state: see git history on branch `fix/systematics-warning`. Fake-data figures: `unfold_output/plot_*_0.pdf` and the per-observable closure dumps `unfold_output/closure_hists_ccpi_Run1_*_xsec.root`. Standalone generator predictions and their build scripts: `generator_predictions/newg4/`. The framework-independent companion analysis (selection, standalone Wiener-SVD / D'Agostini unfolding, and the cross-pipeline validation of §6.7) is documented in `XSecCCPip/analysis_report.md` and `XSecCCPip/report/{analysis,framework}_note.tex`.